Samantha R. Gallatin

Environmental Physics Ph.D. Researcher | UAS Operations | Aerosol Instrumentation | Scientific Python

Tübingen, Germany • samgallatin823@gmail.com • English (native) • German (B2.2)

RESEARCH EXPERIENCE

Ph.D. Researcher – Environmental Physics 2025–Present

University of Tübingen | Tübingen, Germany

• Research atmospheric particles, particle interactions, aerosol measurement, and environmental sensing.
• Operate UAS for scientific data collection under A1/A3 and A2 remote-pilot certifications.
• Plan and coordinate field campaigns, including logistics, operational documentation, instrumentation, and data workflows.
• Contribute to SORA-related documentation and cross-border applications for UAS research.
• Calibrate and validate scientific systems, assess performance, and troubleshoot hardware and software.
• Develop Python scripts and reusable data-processing pipelines for instrument control, quality assurance, visualization, and reproducible scientific analysis.
• Independently designed, implemented, trained, and evaluated an augmented graph neural network (GNN), including model validation, performance analysis, and iterative refinement.
• Teach university courses, present research at conferences, contribute to public outreach, and publish peer-reviewed research.

Student Research Assistant – UAS & Field Campaigns 2024–Jan 2025

University of Tübingen | Tübingen, Germany

• Participated in atmospheric field campaigns and supported scientific UAS data collection.
• Processed, organized, and analyzed UAS-derived environmental datasets.
• Assisted with field preparation, instrument handling, operational workflows, and post-campaign data management.

Graduate Researcher – Renewable Energy 2024

University of Tübingen | Tübingen, Germany

• Built profiles of solar- and wind-energy capacity across Germany.
• Used ERA5 data to investigate energy-storage potential during extreme weather.

Graduate Researcher – Aerosols & Climate 2023

University of Tübingen | Tübingen, Germany

• Collected aerosol-concentration observations and analyzed related climatological conditions.
• Visualized aerosol and climate trends using Python-based time-series analysis.

TECHNICAL EXPERTISE

Programming: Python, R, MATLAB, machine learning (PyTorch and TensorFlow), time-series analysis, data visualization

Atmospheric science: Aerosol measurement, particle interactions, Mie theory, calibration, validation, quality assurance

UAS & fieldwork: A1/A3 and A2 certification, campaign coordination, SORA preparation, cross-border applications

Instrumentation: Scientific sensors, troubleshooting, soldering, Raspberry Pi, basic microelectronics

Geospatial & climate: ArcGIS Pro, ArcMap, GIS, GPS, ERA5, PlaSim

Communication: Conference presentations, teaching, outreach, LaTeX, Overleaf, Beamer

EDUCATION

Ph.D. Environmental Physics: University of Tübingen | Jan 2025–Present

M.Sc. Applied & Environmental Geoscience: University of Tübingen | Aug 2024 | Grade 1.40

B.S. Geography: SIUE | May 2022 | GPA 4.00/4.00

B.S. Biological Sciences: SIUE | May 2022 | GPA 4.00/4.00

EARLIER RESEARCH EXPERIENCE

Undergraduate Researcher – Carbon Monoxide & Severe Weather 2020–2022
Southern Illinois University Edwardsville | Edwardsville, IL, USA
• Created a national dataset of carbon-monoxide deaths and hospitalizations linked to winter and tropical storms using news archives, CDC records, and StormData.
• Used GIS and Excel to analyze spatial and temporal patterns from 1978–2021.
• Presented findings at the Association of American Geographers annual meeting and an undergraduate research showcase.

Undergraduate Researcher – Biology Education 2020
Southern Illinois University Edwardsville | Edwardsville, IL, USA
• Qualitatively coded student responses to assess perceptions of scientific identity.
• Collaborated with a four-person research team to evaluate themes, bias, and uncertainty.

TEACHING & ACADEMIC EXPERIENCE

Atmospheric Physics Tutor 2024
University of Tübingen | Tübingen, Germany
• Guided students through atmospheric-physics exercises and supported seminar presentations.

Academic Student Assistant (HiWi) 2023
University of Tübingen | Tübingen, Germany
• Converted lecture content into publication-ready course scripts and formatted materials in LaTeX and Beamer.

International Tutor – Tübingen START 2023
University of Tübingen | Tübingen, Germany
• Developed orientation resources and supported incoming international students.

Science Tutor & Peer-Led Learning Leader 2019–2022
Southern Illinois University Edwardsville | Edwardsville, IL, USA
• Explained biology, chemistry, and geography concepts and created collaborative learning materials.

SELECTED PUBLICATIONS & PRESENTATIONS

• Gallatin, S., Savvakis, V., Schön, M., Kezoudi, M., Lekaki, N., Papetta, A., et al. (2026). Impact of refractive index corrections of the OPC-N3 on Saharan dust aerosol particle size distributions. Aerosol Science and Technology, 1–25.
• Gallatin, S.R. & Black, A.W. (2022). “The Silent Killer of Winter Storms: Carbon Monoxide Poisoning Across the United States from 1978–2021.” Association of American Geographers Annual Meeting.
• Gallatin, S.R. (2022). “Carbon Monoxide Poisoning in Tropical and Winter Storms Across the United States: A Comparison of Deaths and Injuries.” SIUE Undergraduate Student Showcase.

LEADERSHIP, SERVICE & HONORS

Leadership: Study Commission Student Representative, University of Tübingen (2023–2024)

Leadership: Co-President and Secretary, Wildlife and Conservation Biology Club, SIUE (2021–2022)

Service: Environmental, animal-welfare, and food-access volunteering

Awards: Distinguished Student of Geography Award (2022)

Awards: Senior Award in Genetics and Cell Biology (2022); Lincoln Laureate of SIUE (2021)

Scholarships: Meridian Scholar – Full Tuition Scholarship; Dean’s List (2018–2021)